

**Question Set 6**

Comments: some questions in the question set are not like exam or test questions. They are used for the purpose of reflecting on some topic to ensure students understand the main concepts. These will be called here "Reflecting questions" (RQ). The questions more similar to exam/test questions will be labelled ETQ.

**In class questions**

- 1) Have a look at the Lecture Notes Set 3 (Slides 38-47 **only**) and ask questions of what you do not understand.
- 2) Explain why polyethylene synthesised by free-radical polymerisation is not a high-performance material. (You may need to find out what that means). Use SPRE in your answer.

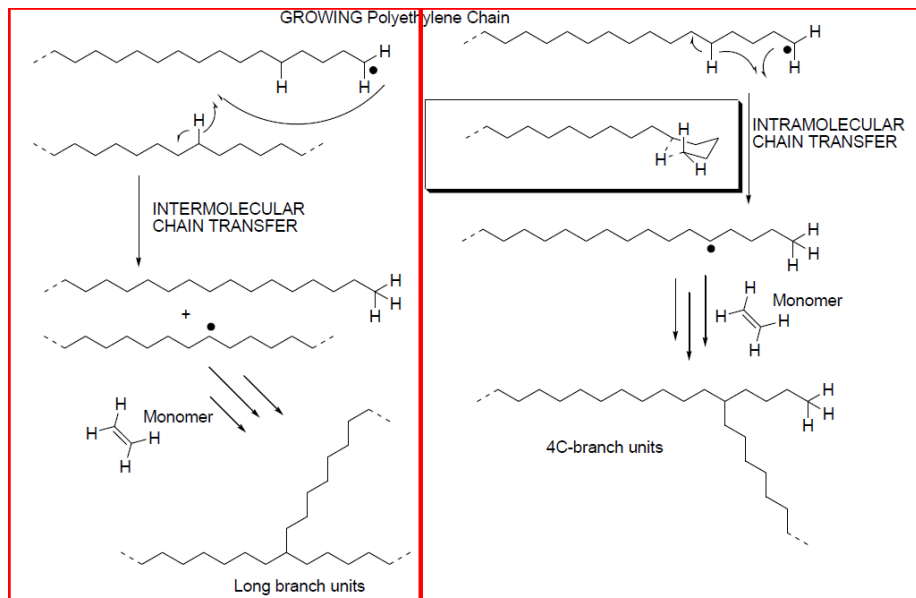
ETQ: For full answers you need a SPRE style answer (but you don't need to label S-P-R-E, I will do it here so that you understand what they are) AND show structures and mechanisms.

S: High Performance Materials are materials with excellent thermomechanical properties. For this, you need polymers that pack well, with low free-volume, relatively rigid, and for strong fibres, with high levels of crystallinity.

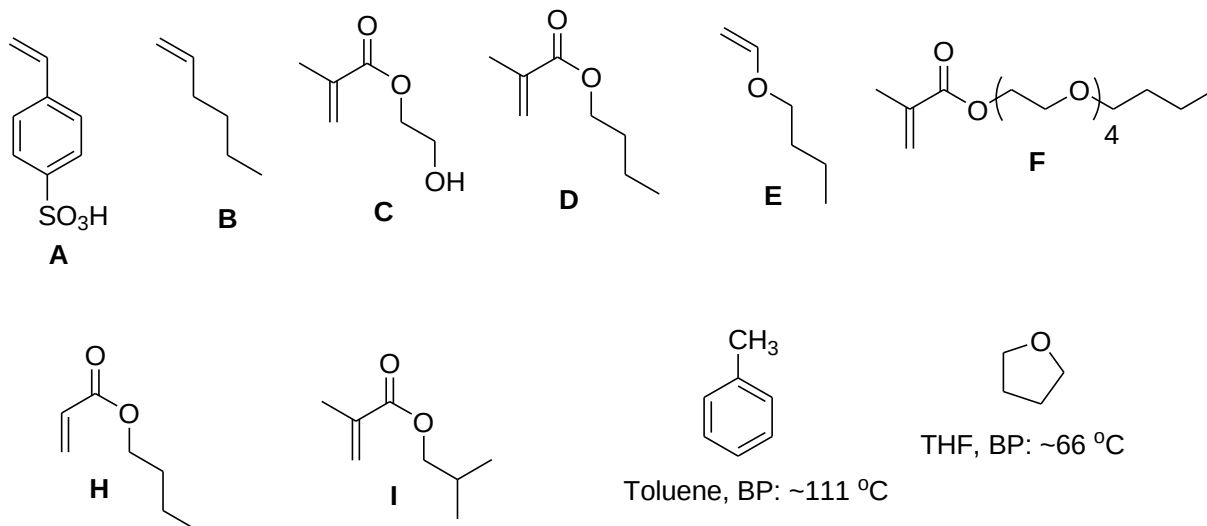
P: for a polymer like PE, that only has CH<sub>2</sub>-CH<sub>2</sub> bonds, no side groups, therefore, no structural source of H-bonding, high polarity or pi-pi stacking, achieving high-performance, means excellent packing, fully linear and high MW. SO, why would PE obtained by FRP not comply?

R: because of chain transfer. Chain transfer to polymer introduces a great deal of branching, which makes the polymer increase free-volume and make it a very amorphous material, with little packing.

E: The mechanisms in the Lecture Notes show this:



- 3) You are instructed to synthesise a **hydrophobic** polymer. You are provided with the monomers, solvents and reagents shown below from commercial sources. You can have any free-radical initiator from those who appear in the Lecture Notes.



- a) Which monomer/s will you choose and why? Explain briefly why you would discard the others.

The monomers chosen need to fulfill two separate criteria: 1-be able to form a hydrophobic monomer 2-polymerise through FRP.

A: FRP? Yes (styrene) / Hydrophobic? No (sulphonic acid side chain: VERY polar + H-bonding)  
 B: FRP? No (olefin: chain transfer to monomer, like seen in class for PP) / Hydrophobic? Yes  
 C: FRP? Yes (methacrylate) / Hydrophobic? No (OH side chain: VERY polar + H-bonding)  
 D: FRP? Yes (methacrylate) / Hydrophobic? YES  
 E: FRP? No (vinyl ether: chain transfer to monomer, like seen in class) / Hydrophobic? Yes  
 F: FRP? Yes (methacrylate) / Hydrophobic? No (oligo ethylene glycol side chain: VERY polar once all side chains have this functionality)  
 H: FRP? Yes (acrylate) / Hydrophobic? YES  
 I: FRP? Yes (methacrylate) / Hydrophobic? YES

Discarded for no FRP: B, E

Discarded for hydrophilic: A, C, F

Proposed: D, H, I

- b) In addition, you are further requested that your polymer has the highest T<sub>g</sub> possible. How will that affect your choice in a) and which monomer will you choose now?

Acrylates have lower T<sub>g</sub> than methacrylates because of the missing CH<sub>3</sub> (CH<sub>3</sub> restricts rotational mobility of main chain, therefore, higher T<sub>g</sub> for methacrylates). This means H lower T<sub>g</sub> than D and I. Likely that D has highest T<sub>g</sub> because I has a branching unit slightly away from the main chain, therefore possibly affecting other chains more than the same chain, increasing free-volume and therefore, facilitating rotation of the main chain, decreasing T<sub>g</sub>.

- c) Prior to the free-radical polymerisation, you would purify the commercially available monomer chosen in b), even if high purities are commercially available. Explain why this is normally the case.

See Question Set 5. This is explained in Question 2.

- d) Select a suitable free-radical initiator to use if THF is used as solvent, and another one if toluene is used as solvent. Explain your choice.

Normally, reactions are heated at near the boiling point of the solvent (you cannot heat too much higher, as the solvent boils at this temperature). Therefore, you try to use initiators with decomposition temperatures closer to the boiling point of the solvent. AIBN and azo initiators have decomposition temperatures close to 70 °C, and peroxy initiators close to 110-120 °C. Therefore, I would choose AIBN for THF and BPO, for example, for toluene.

- e) Do you expect the polymer to have higher molecular weight if the monomer was polymerised using THF or toluene as solvents? Explain. What parameter(s) would you need to know exactly to determine this in a quantitative way?

THF was shown in the Lecture Notes as an example of solvent acting as CTA (with maleimides). Of course, this behavior would be monomer-dependent, but I would avoid THF in principle, if this was possible, as THF will likely result in shorter chains, if chain-transfer to solvent is confirmed for these monomers. You need the  $k_{tr}$  or the  $C_T$  for the solvent/monomer pairs.

- f) If you wanted to avoid the issues discussed in e), would you simply avoid the use of solvents? Discuss.

Possibly, if the monomer was a liquid (therefore, acting as a solvent itself). However, other problems may appear, for example, in bulk polymerisations (no solvent), gel-effect, a deviation from steady-state kinetics, is normally more prominent because of much higher viscosity in the absence of solvent.